



Competition and the Phillips curve[☆]

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ABSTRACT

We propose a mechanism through which the rising market concentration causes the flattening of the Phillips curve. By extending the New Keynesian model using Homothetic Single Aggregator (HSA) demand systems, we show that a higher entry cost reduces the number of firms, which raises the markup rate and lowers the pass-through rate, thereby structurally flattening the Phillips curve. Additionally, an endogenous entry creates cost-push shocks through strategic complementarity, biasing empirical estimates. These two reasons, one structural and one observational, together can go a long way toward understanding the flattening of the Phillips curve.

1. Introduction

One of the major issues central banks have been facing in recent years is the flattening of the Phillips curve. For example, Federal Reserve Vice Chair, Richard Clarida said on Sept. 26, 2019, that “Another key development in recent decades is that price inflation appears less responsive to resource slack. That is, the short-run price Phillips curve – if not the wage Phillips curve – appears to have flattened.” San Francisco Fed President, Mary Daly stated on Aug. 29, 2019, that “As for the Phillips curve... most arguments today center around whether it’s dead or just gravely ill”. New York Fed President, John Williams explained on Feb. 22, 2019, that “The Phillips curve is the connective tissue between the Federal Reserve’s dual mandate goals of maximum employment and price stability. Despite regular declarations of its demise, the Phillips curve has endured. It is useful, both as an empirical basis for

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forecasting and for monetary policy analysis.” Central banks are wary of it since achieving price stability becomes more challenging due to the reduced impact of monetary policy on inflation rates through real economic activities.

Another significant development that the U.S. economy has experienced in recent years is market concentration. There is a growing concern about the adverse impacts of the resulting increase in profit margins and markups on the macroeconomy. Recent studies provide empirical evidence of increasing market concentration. Covarrubias et al. (2019) show that “After 2000, however, the evidence suggests inefficient concentration, decreasing competition and increasing barriers to entry, as leaders become more entrenched and concentration is associated with lower investment, higher prices and lower productivity growth.” According to De Loecker et al. (2020), “In 1980, aggregate markups start to rise from 21% above marginal cost to 61% now. ... We also find an increase in the average profit rate from 1% to 8%. Although there is also an increase in overhead costs, the markup increase is in excess of overhead.” In addition, Autor et al. (2020) state that “sales concentration is rising across a large set of industries. ... aggregate markups have been rising.”

These two events observed in recent years seem unrelated according to the textbook New Keynesian model (see, for example, Gali, 2015, Walsh, 2010 and Woodford, 2003). Under the CES demand system extensively used in the literature, market concentration neither leads to the flattening of the Phillips curve nor directly impacts the inflation rate through the Phillips curve. This irrelevance result can be overturned once we depart from CES and adopt more flexible demand systems. The contribution of this paper is to offer a theoretical framework that reveals the role of market concentration on the flattening of the Phillips curve.

Obviously, neither market concentration nor the slope of the Phillips Curve should be treated as exogenous parameters. Accordingly, we develop a model in which both of them are endogenously determined by underlying exogenous factors. In particular, we examine a scenario in which an increase in entry costs leads to greater market concentration and a flattening of the Phillips Curve. More specifically, we extend the canonical New Keynesian model under CES monopolistic competition in two respects. First, we incorporate entry and exit as in Bilbiie et al. (2008) and Bilbiie et al. (2014). Second, we replace CES by *Homothetic Single Aggregator* (HSA), a class of homothetic demand systems, originally proposed in a general setting by Matsuyama and Ushchev (2017) and subsequently extended to monopolistic competition by Matsuyama and Ushchev (2020b). We use HSA for several reasons:

- HSA contains CES and Translog as special cases.
- It is tractable due to its single aggregator that serves as a sufficient statistic, because it summarizes all the impacts of competitive pressures on the pricing behavior of monopolistically competitive firms, including the markup rate and the pass-through rate.
- HSA is flexible. For example, HSA can accommodate *Marshall's second law of demand* (hereafter, the Second law)¹ as well as what Matsuyama and Ushchev (2023b) call *the third law of demand* (hereafter, the Third law).²
- HSA is restrictive enough to ensure that the equilibrium exists uniquely even under endogenous entry and that the Second law guarantees *procompetitive entry*, i.e., entry of new firms reduces equilibrium markup rate.³

These features of HSA enable us to show how the entry affects the Phillips curve through its impact on the price elasticity and the pass-through rate. To the best of our knowledge, this paper is the first attempt to incorporate the HSA demand system into the New Keynesian framework.

For price adjustment mechanisms, we use both Rotemberg (1982) and Calvo (1983) because they are no longer equivalent under the HSA demand system. Under Rotemberg pricing, the slope of the New Keynesian Phillips curve (hereafter, NKPC) depends on the price elasticity, but not on the pass-through rate, so that the Second law, not the Third law, plays the crucial role in the flattening of NKPC. In contrast, under Calvo pricing, it depends on the pass-through rate, but not on the price elasticity, so that the Third law, not the Second law, plays the crucial role.⁴

Here are our main theoretical findings. Under the Second and the Third laws, a higher entry cost, which leads to market concentration, causes the flattening of the Phillips curve for two reasons. First, market concentration, which leads to a decline of the price elasticity under the Second law and of the pass-through rate under the Third law, causes a *structural* flattening of the Phillips curve. This is what we call the *steady-state effect of concentration*. Second, an endogenous entry generates cost-push shocks through strategic complementarity, which we call *the dynamic effect of endogenous entry*. Due to this endogeneity, a *naïve* regression of the inflation rate on the real marginal cost leads to the negative omitted variables bias under the Second law, whose magnitude is amplified with more concentration under the Third law (in the case of Rotemberg) and under both the Second and the Third laws

¹ The Second law states that the price elasticity of demand goes up with its price, which implies that market concentration leads to a higher markup rate in our setup and that more productive firms have higher markup rates with heterogeneous firms as in Matsuyama and Ushchev (2023b). This is in line with the empirical evidence by Campbell and Hopenhayn (2005), Burstein and Gopinath (2014), De Loecker and Goldberg (2014), Feenstra and Weinstein (2017), Amiti et al. (2019) and Baqaee et al. (2023).

² The Third law in its strong (weak) form states that the rate of increase in the price elasticity goes down (does not go up) with its price, which implies that market concentration leads to a lower (does not lead to a higher) pass-through rate in our setup and that more productive firms have lower (do not have higher) pass-through rates with heterogeneous firms in Matsuyama and Ushchev (2023b). This is in line with the empirical evidence by Berman et al. (2012), Amiti et al. (2014) and Baqaee et al. (2023).

³ As explained in Matsuyama (2025), there may be multiple equilibria under general homothetic demand systems. Moreover, contrary to the widely-held belief, referred to as “Fallacy 4” in Matsuyama (2025), the Second law is neither sufficient nor necessary for procompetitive entry in general.

⁴ We conjecture that both the Second and the Third laws matter and that results would be a hybrid of Rotemberg and Calvo under more general price adjustment mechanisms.

(in the case of Calvo).⁵ This weakens the *estimated* relationship between the inflation rate and the marginal cost. These two reasons, one structural and one observational, together can go a long way toward understanding the flattening of the Phillips curve.

Our simulation suggests that market concentration generates a sizable reduction in the slope of NKPC. We show that the rise in the markup rate as observed in the data can halve the slope of NKPC. As a result, in impulse responses, the responses of the inflation rate to both technology and monetary policy shocks become smaller as market concentration deepens.

It should be also pointed out that under the Second and the Third laws, the cyclicalities of the markup rate is determined by the tension between nominal rigidities and the pass-through rate. Suppose a positive technology shock hits the economy. As is well-known, in a sticky price equilibrium under the CES demand system, the markup rate is procyclical, because the price does not respond in the short run to the decline in the marginal cost. But, under the Second law, a positive technology shock, which increases the number of firms, works in the direction of reducing the markup rate and making it countercyclical in a flexible price equilibrium. This explains why the cyclicalities of the markup rate in a sticky price equilibrium under the Second and the Third laws is generally ambiguous.⁶

1.1. Literature

There is a vast literature on the flattening of the Phillips curve, which can be divided into five categories. First, Goolsbee and Klenow (2018) and Crump et al. (2019) attribute this to mis-measurements in the inflation rate and gap measures. Second, Daly and Hobijn (2014) argue that there has been a structural change in the labor market and that the flattening happened not to the price Phillips curve but to the wage Phillips curve. Third, McLeay and Tenreiro (2019) argue that changes in monetary policy response to inflation have weakened the observed relationship between the gap measures and the inflation rate. Fourth, Coibion and Gorodnichenko (2015) and Hazell et al. (2022) state that the declining slope coefficient is due to the declining inflation expectations not being properly taken into account when estimating the Phillips curve. Fifth, Sbordone (2010), Shirota (2015), Wang and Werning (2022), Baqaee et al. (2024), Harding et al. (2022), L'Huillier et al. (2022) and Rubbo (2023) propose models in which the slope of NKPC becomes structurally flatter in line with the empirical evidence in Del Negro et al. (2020). Our paper is related both to the fourth because of the omitted variable bias result, and to the fifth because of the structural flattening result.

It is well-known that in the prototypical New Keynesian model under the CES demand system, competition is irrelevant to the Phillips curve, irrespective of entry and exit. It neither changes the slope of the Phillips curve nor exert any direct impact on the inflation rate. A few recent studies challenge this irrelevance of competition.⁷ Bilbiie et al. (2008), even though its primary focus is not on the causality from market concentration and the flattening of the Phillips curve, points out that, under Rotemberg using Translog, the slope of the Phillips curve is an increasing function of the number of firms and that an endogenous cost-push shock emerges.⁸ Wang and Werning (2022) show that in a dynamic oligopoly model with strategic interaction, higher concentration leads to amplified real effects of monetary policy, the Phillips curve with inflation persistence, and the endogenous cost-push shock. Baqaee et al. (2024) demonstrate that higher concentration leads to the flattening of the Phillips curve through two different channels: the real rigidities and the misallocation across heterogeneous firms. They call the second channel the supply side effects of monetary policy, because monetary easing raises aggregate TFP by shifting resources to more efficient firms. Our model also has the supply side effects of monetary policy, but the mechanism is different. Our mechanism operates through the combination of endogenous entry of firms and non-CES. We use HSA for non-CES, because of its analytical tractability, instead of extending the Kimball (1995) aggregator to allow for endogenous entry.⁹ Table 1 highlights the differences among these two and our studies.

Our paper is also related to monopolistic competition models with entry and exit. Bilbiie et al. (2012, 2019) construct the flexible price model while Bilbiie et al. (2008) and Bilbiie et al. (2014) set up the sticky price model. Bilbiie et al. (2012) shows that the endogenous entry models can account for business cycles as well as the standard RBC models. Bilbiie et al. (2014) and Bilbiie et al. (2019) characterize the optimal policy under entry and exit. Although they offer some robustness checks using a few parametric non-CES demand systems including Translog, the endogeneity of markup and pass-through rates does not play the central role in these four studies. In contrast, our study reveals crucial roles of the endogeneity of markup and pass-through rates with endogenous entry using nonparametric HSA.

Our paper also contributes to the debate on the equivalence/nonequivalence between Calvo and Rotemberg pricing. Under CES, both result in a common formulation of NKPC to the first order approximation, as shown by Roberts (1995). Nistico (2007) shows that these two price settings yield the same welfare losses to a second order of approximation around the efficient steady state,

⁵ Bilbiie et al. (2008) also show that an endogenous cost-push term emerges under Rotemberg using Translog, and point out that failing to account for this term when estimating the Phillips curve introduces bias.

⁶ There is indeed a disagreement on the cyclicalities of the markup rate in the literature as surveyed by Nekarda and Ramey (2020).

⁷ A few early studies hint the possible structural relationship between competition and the Phillips curve. Means (1936) states that “to a major extent, technology and economic concentration have brought a change in the demand curve faced by the individual producer. ... In the industries dominated by a few big competitors, administered prices would undoubtedly appear and though there might be sufficient competition to make them reasonably fair, they would be likely to be inflexible.” Carlton (1986) states, “The level of industry concentration is strongly correlated with rigid prices. The more concentrated the industry, the longer is the average spell of price rigidity.”

⁸ Incidentally, Translog violates the Third law and hence implies that, under Calvo, market concentration makes the Phillips curve steeper contrary to the empirical evidence.

⁹ The Kimball aggregator is the most popular choice for non-CES in the macro literature, such as Dotsey and King (2005), Klenow and Willis (2016), Harding et al. (2022), Baqaee et al. (2024) and Aruoba et al. (2024). However, extending Kimball to allow for endogenous entry is a challenge. Perhaps for this reason, the existing macro models with Kimball assume a fixed set of firms without entry.

Table 1
Monetary policy and competition.

	Comp.	Entry	Pref.	Nominal friction
Bilbiie et al. (2008)	mono.	end.	CES/Translog	Rotemberg
Wang and Werning (2022)	oligo.	exo.	CES/Kimball	Calvo
Baqae et al. (2024)	mono.	exo.	Kimball	Calvo
our paper	mono.	end.	HSA	Rotemberg/Calvo

oligo.: oligopolistic competition, mono.: monopolistic competition, exo.: exogenous entry, end.: endogenous entry

while Lombardo and Vestin (2008) show that they may entail different welfare costs at higher order approximation around the inefficient steady state. Ascari and Rossi (2012) provides that the long-run Phillips curve, *i.e.*, the long-run relationship between inflation and output, and the dynamics in the presence of the trend inflation are different between these two price settings. Some use the equivalence between Calvo and Rotemberg to argue in favor of Rotemberg because of its analytical simplicity and state dependency, while others use it to argue in favor of Calvo because of its consistency with micro evidence of the size and frequency of individual price changes. Our paper demonstrates that the equivalence between Calvo and Rotemberg fails to hold even as the first order approximation, which suggests both Calvo and Rotemberg offer valuable insights.

The core ingredient of this paper is HSA. Since HSA was proposed by Matsuyama and Ushchev (2017), a growing number of studies have been applying it to explain a variety of issues. Matsuyama and Ushchev (2020b) investigate how the condition for procompetitive vs. anticompetitive entry – the markup rate either goes down or goes up as more firms enter – is related to the condition for excessive vs. insufficient entry. In order to understand how competitive pressures affect selection and sorting of firms with heterogeneous productivity, Matsuyama and Ushchev (2023b) extend Melitz (2003) model to incorporate the endogenous markup embedded in the HSA demand system. Matsuyama and Ushchev (2022) extend Judd (1985) model of endogenous innovation cycles by replacing CES with HSA to show how an increase in market size has the destabilizing effects on the dynamics of innovation. Grossman et al. (2023) discuss whether the policy should promote diversification or re-shoring in the face of insecure supply chains and seek for the policy instruments to achieve efficient supply chains under the Second law. Baqaee et al. (2023) study how an increase in market size, say, due to globalization, affects welfare in a monopolistically competition model with heterogeneous markups and endogenous entry. Trottner (2023) and Ren and Zhang (2025) extend this line of inquiry to the two-sided market-power by using HSA both in the product and labor markets. Kasahara and Sugita (2020) propose a nonparametric estimation approach of the HSA demand system. Our study is the first to apply the HSA demand system to the New Keynesian model, the workhorse model for policy simulations and forecasting in central banks and international organizations.

1.2. Layout

The structure of this article is as follows. Section 2 formally defines the HSA demand system and explains its key properties. Section 3 introduces the New Keynesian model with Rotemberg (1982) pricing under the HSA demand system. Section 4 discusses the relationship between competition and the Phillips curve. Section 5 and Appendix H provide the simulation analysis using parametric families of HSA: CES, Translog and Co-PaTh.¹⁰ Section 5 shows how higher entry cost leads to concentration and the flattening of the Phillips curve, while Appendix H investigates impulse responses to technology and monetary policy shocks by different pass-through rate and by different entry cost. Section 6 studies the relationship between competition and the Phillips curve under Calvo (1983) pricing. There, we explain the reason for non-equivalence between Rotemberg and Calvo. We also show simulation results using another parametric family of HSA called PEM.¹¹ Section 7 concludes. All technical materials are in Appendices.

2. HSA

This section explains the HSA demand system, originally proposed by Matsuyama and Ushchev (2017), which is in turn restricted by Matsuyama and Ushchev (2020b, 2022, 2023b) to a continuum of varieties ($\omega \in \Omega_t$), gross substitutes, and symmetry to be applied for monopolistic competition. See also Matsuyama and Ushchev (2023a).

Consider the single final good, which is produced competitively by assembling differentiated intermediate inputs with the constant-returns-to-scale (CRS) technology, characterized by the unit cost function $P(\mathbf{p}_t)$, where $\mathbf{p}_t$ is the vector of intermediate inputs prices. We call the demand system for intermediate inputs by the final goods producers HSA if the market share for the input variety ω depends solely on its single *normalized price*, *i.e.*, its own price divided by the single price aggregator $A(\mathbf{p}_t)$, which captures all the cross-price effects in the demand system:

$$\frac{\partial \ln(P(\mathbf{p}_t))}{\partial \ln(p_t(\omega))} = \frac{p_t(\omega) c_t(\omega)}{P(\mathbf{p}_t) C_t} = s \left(\frac{p_t(\omega)}{A(\mathbf{p}_t)} \right), \quad (1)$$

¹⁰ Co-PaTh, proposed by Matsuyama and Ushchev (2020a), stands for Constant Pass-Through. It satisfies the Second law and the Third law in the weak form, and contains CES as a limit case.

¹¹ PEM, proposed by Matsuyama and Ushchev (2023b), stands for Power Elasticity of Markup rate. It satisfies the Third law in the strong form and contains Co-PaTh as a limit case, which in turn contains CES as a limit case.

where

$$\int_{\Omega_t} s \left(\frac{p_t(\omega)}{A(\mathbf{p}_t)} \right) d\omega \equiv 1. \quad (2)$$

$s : \mathbb{R}_{++} \rightarrow \mathbb{R}_+$ is the market share function, decreasing in the normalized price $z_t(\omega) := p_t(\omega)/A(\mathbf{p}_t)$ for $s(z) > 0$ with $\lim_{z \rightarrow \bar{z}} s(z) = 0$. If $\bar{z} := \inf \{z > 0 | s(z) = 0\} < \infty$, $\bar{z}A(\mathbf{p}_t)$ is the choke price. Because varieties are gross substitutes, the market share is decreasing in z : $s'(z) < 0$.¹²

The single price aggregator $A_t = A(\mathbf{p}_t)$ is implicitly defined by the adding-up constraint in Eq. (2). By construction, market shares add up to one. As is evident from Eq. (2), $A(\mathbf{p}_t)$ is linear homogeneous in $\mathbf{p}_t$ for fixed Ω_t . A larger Ω_t , namely, more variety, reduces $A(\mathbf{p}_t)$.

By integrating Eq. (1), one can show that $A(\mathbf{p}_t)$ and $P(\mathbf{p}_t)$ are related as follows:

$$\frac{P(\mathbf{p}_t)}{A(\mathbf{p}_t)} = \exp \left\{ \bar{K} - \int_{\Omega_t} \left[\int_{\frac{p_t(\omega)}{A(\mathbf{p}_t)}}^{\bar{z}} \frac{s(\xi)}{\xi} d\xi \right] d\omega \right\}, \quad (3)$$

where $\bar{K}$ is a constant.¹³

$A(\mathbf{p}_t)$ is the inverse measure of *competitive pressures* and fully captures the cross-price effects in the demand system. It is different from the unit cost function of the final goods $P_t = P(\mathbf{p}_t)$, which is the inverse measure of TFP and captures the *productivity consequences* of price changes. In general, they do not move together: $A(\mathbf{p}_t) \neq \text{constant} \times P(\mathbf{p}_t)$. This can be seen by differentiating the adding-up constraint in Eq. (2):

$$\frac{\partial \ln(A(\mathbf{p}_t))}{\partial \ln(p_t(\omega))} = \frac{\frac{p_t(\omega)}{A(\mathbf{p}_t)} s' \left(\frac{p_t(\omega)}{A(\mathbf{p}_t)} \right)}{\int_{\Omega_t} \frac{p_t(\omega')}{A(\mathbf{p}_t)} s' \left(\frac{p_t(\omega')}{A(\mathbf{p}_t)} \right) d\omega'} \neq s \left(\frac{p_t(\omega)}{A(\mathbf{p}_t)} \right) = \frac{\partial \ln(P(\mathbf{p}_t))}{\partial \ln(p_t(\omega))},$$

which shows that $A(\mathbf{p}_t) \neq \text{constant} \times P(\mathbf{p}_t)$ unless $zs'(z)/s(z) < 0$ is constant, in other words, $s(z)$ is a decreasing power function. This corresponds to CES because plugging $s(z) = \gamma_{CES} z^{1-\theta}$, where $\theta > 1$, into Eqs. (1) to (3) can verify

$$C_t = Z_C \left[\int_{\Omega_t} c_t(\omega)^{1-\frac{1}{\theta}} d\omega \right]^{\frac{\theta}{\theta-1}}, \quad (4)$$

$$c_t(\omega) = Z_C^{\theta-1} \left(\frac{p_t(\omega)}{P(\mathbf{p}_t)} \right)^{-\theta} C_t,$$

$$Z_C^{\theta-1} \left(\frac{p_t(\omega)}{P(\mathbf{p}_t)} \right)^{1-\theta} = \frac{p_t(\omega) c_t(\omega)}{P(\mathbf{p}_t) C_t} = s \left(\frac{p_t(\omega)}{A(\mathbf{p}_t)} \right) = \gamma_{CES} \left(\frac{p_t(\omega)}{A(\mathbf{p}_t)} \right)^{1-\theta},$$

where Z_C is the TFP of the final goods production, and related to $\bar{K}$ and γ_{CES} as follows:

$$\frac{P(\mathbf{p}_t)}{A(\mathbf{p}_t)} = \exp \left(\bar{K} - \frac{1}{\theta-1} \right) = \frac{\gamma_{CES}^{\frac{1}{\theta-1}}}{Z_C}. \quad (5)$$

Three price indices. We consider endogenous entry (variety). Even under symmetric situation, individual price is not equal to the aggregate price due to entry effects. In addition, we have two aggregate prices: the final goods price and the single price aggregate. As a result, we need to distinguish three separate aggregate prices.

The first one is the final goods price or CPI P_t , which captures the productivity effects of entry, and is the reference price for consumers, implicitly given by

$$\int_{\Omega_t} \frac{p_t(\omega) c_t(\omega)}{P_t C_t} d\omega \equiv 1.$$

The second is the single price aggregate A_t , which captures the competitive effects of entry, and is the reference price for firms, implicitly given by

$$\int_{\Omega_t} s \left(\frac{p_t(\omega)}{A_t} \right) d\omega \equiv 1.$$

The third one is the average price index or PPI p_t :

$$p_t = \int_{\Omega_t} s \left(\frac{p_t(\omega)}{A_t} \right) p_t(\omega) d\omega.$$

¹² Proposition 1 in Matsuyama and Ushchev (2017) proves the existence of well-defined CRS production technologies that rationalize the HSA demand system as described above.

¹³ For the detailed derivation of Eq. (3), see Matsuyama and Ushchev (2017).

Unlike P_t and A_t , the average price index p_t is not affected by entry effects and therefore is the measured price index. Thus, in what follows, we consider the inflation rate in terms of p_t when evaluating the responsiveness of inflation rates to macroeconomics variables in NKPC.

3. New Keynesian model under HSA

Our model is an extension of those in Bilbiie et al. (2008) and Bilbiie et al. (2014) to incorporate the HSA demand system. We consider a closed economy populated by four agents: the representative household, competitive final goods producers, monopolistically competitive intermediate inputs producers, and the central bank. Intermediate inputs producers set prices subject to nominal rigidities. Their entry is endogenous but they are subject to exogenous exit shocks.

Timing. Time is discrete: $t = 0, 1, 2, 3, \dots$. There is an unbounded mass of potential entrants in every period. They are subject to one-period time-to-build lag. That is, entrants at time t only start producing at time $t + 1$. Entry is determined endogenously by the free entry condition, but exit is exogenous. All firms face the same probability δ of exogenous firm destruction at the end of each period, after production and entry. As a result, a proportion δ of new entrants never produces.

3.1. Household

The representative household chooses consumption C_t , labor supply L_t , the nominal bond B_t , and the equity of intermediate inputs producer x_t , in order to maximize welfare:

$$\mathbb{E}_t \sum_{i=0}^{\infty} \beta^i (u(C_t) - v(L_t)), \quad (6)$$

subject to the budget constraint:

$$\frac{B_{t+1}}{P_t} + x_{t+1} \int_{\Omega_t + \Omega_{E,t}} \frac{V_t(\omega)}{P_t} d\omega + C_t = (1 + i_{t-1}) \frac{B_t}{P_t} + x_t \int_{\Omega_t} \frac{D_t(\omega) + V_t(\omega)}{P_t} d\omega + \frac{W_t}{P_t} L_t. \quad (7)$$

where β denotes the subjective discount factor and δ the probability of exogenous termination. W_t denotes the nominal wage and i_t the nominal interest rate. $V_t(\omega)$ and $D_t(\omega)$ denote the equity price (the value) and the profit of the intermediate inputs producer $\omega \in \Omega_t$.

3.2. Final goods producers

Final goods producers are under perfect competition and produce the final good by assembling intermediate inputs using the CRS technology that generates the HSA demand system in Eqs. (1) and (2).

3.3. Intermediate inputs producers

Intermediate inputs producer ω uses labor $l_t(\omega)$ to produce its output $y_t(\omega)$ with the linear technology:

$$y_t(\omega) = Z_{P,t} l_t(\omega), \quad (8)$$

where $Z_{P,t}$ is the common technology in production. Intermediate inputs producer ω chooses the price path $p_t(\omega)$ to maximize the value of the firm:

$$\frac{V_t(\omega)}{P_t} = \mathbb{E}_t \sum_{i=1}^{\infty} [\beta(1 - \delta)]^i \frac{u'(C_{t+i})}{u'(C_t)} \left(\frac{D_{t+i}(\omega)}{P_{t+i}} \right), \quad (9)$$

subject to the HSA demand curve in Eq. (1):

$$y_t(\omega) = c_t(\omega) = s \left(\frac{p_t(\omega)}{A_t} \right) \frac{P_t C_t}{p_t(\omega)}, \quad (10)$$

and the Rotemberg (1982) quadratic price adjustment cost¹⁴:

$$\frac{D_t(\omega)}{P_t} = \frac{p_t(\omega)}{P_t} y_t(\omega) - \frac{W_t}{P_t} l_t(\omega) - \frac{\chi}{2} \left(\frac{p_t(\omega)}{p_{t-1}(\omega)} - 1 \right)^2 \frac{p_t(\omega)}{P_t} y_t(\omega), \quad (11)$$

where χ scales the size of the cost.

We use Rotemberg (1982) pricing to highlight the role of the Second law until Section 5. In Section 6, we consider Calvo (1983) pricing to highlight the role of the Third law.

In this model, we depart from CES within HSA only in the direction of the Second law. As shown in Matsuyama and Ushchev (2020b, Proposition 1) and Matsuyama (2025, Section 6), this guarantees that the equilibrium is unique and symmetric.

¹⁴ For simplicity, we follow Bilbiie et al. (2008) and Bilbiie et al. (2014) to assume that new entrants also incur Rotemberg adjustment costs based on the prices set by incumbents in the previous period. In Appendix B, we derive the NKPC under the scenario where entrants have the flexibility to set prices. Main results carry over under this alternative scenario.

3.4. Aggregate conditions and others

Monetary policy. The central bank sets the nominal interest rate following the simple feedback rule reacting to the PPI inflation rate:

$$(1 + i_t) = (1 + i_{t-1})^{\alpha_i} \left(\frac{p_t}{p_{t-1}} - 1 \right)^{(1-\alpha_i)\alpha_\pi} u_t, \quad (12)$$

where $\alpha_\pi > 1$ to satisfy the Taylor principle. α_i represents the policy inertia. u_t denotes the monetary policy shock.

Free entry. Entrants pay the entry cost $W_t/Z_{E,t}f_{E,t}$, where $Z_{E,t}$ and $f_{E,t}$ are the common technology to facilitate entry and the labor demand for entry purpose, respectively. Since the equilibrium is symmetric, entry occurs until the firm value becomes equal to the entry cost, resulting in the free entry condition:

$$\frac{W_t}{P_t} \frac{f_{E,t}}{Z_{E,t}} = \frac{V_t}{P_t}. \quad (13)$$

Firm dynamics.

$$N_t = (1 - \delta) (N_{t-1} + N_{E,t-1}). \quad (14)$$

N_t is the number of firms (varieties) and the mass of Ω_t . $N_{E,t}$ is the number of entrants and the mass of $\Omega_{E,t}$.

Market clearing. Financial market clearing conditions for the nominal bond and the equity are given by $B_t = 0$ and $x_t = 1$, respectively.

The labor market clearing condition is given by

$$L_t = N_t l_t + N_{E,t} \frac{f_{E,t}}{Z_{E,t}}. \quad (15)$$

The left hand side is the supply of the labor. The first and second terms on the right hand side are the labor demands for production and entry, respectively.

Inserting Eq. (14) into the budget constraint in Eq. (7) leads to

$$\frac{V_t}{P_t} N_{E,t} + C_t = \frac{D_t}{P_t} N_t + \frac{W_t}{P_t} L_t.$$

Inserting Eqs. (11), (13) and (15) into the above aggregate accounting equation yields the resource constraint:

$$C_t = \left[1 - \frac{\chi}{2} (\pi_t - 1)^2 \right] N_t \frac{p_t}{P_t} y_t = \left[1 - \frac{\chi}{2} (\pi_t - 1)^2 \right] Y_t.$$

Output is either consumed or used for the price adjustment costs.

HSA demand system. As the equilibrium is symmetric, the market share is simply expressed by the inverse of the number of firms:

$$s \left(\frac{p_t}{A_t} \right) = \frac{1}{N_t}. \quad (16)$$

The equation for the reference prices for households and intermediate inputs producers in Eq. (3) simplifies to

$$\ln \left(\frac{p_t}{A_t} \right) = \bar{K} - \frac{1}{s \left(\frac{p_t}{A_t} \right)} \int_{\Omega_t} \left[\int_{\frac{p_t(\omega)}{A_t}}^{\bar{z}} \frac{s(\xi)}{\xi} d\xi \right] d\omega, \quad (17)$$

where the second term on RHS captures the love-for-variety effect in the HSA demand system.¹⁵

3.5. Equilibrium

Definition. An *equilibrium* in this economy is a collection of sequence of aggregate prices $\{P_t, A_t, W_t, i_t\}$ and the price of intermediate goods $\{p_t\}$; a collection of sequences of aggregate quantities $\{Y_t, C_t, L_t\}$ and quantities of intermediate goods $\{y_t, l_t\}$; and a collection of sequences of firm-value functions and profit $\{V_t, D_t\}$ together with measures of operating firms and entering firms $\{N_t, N_{E,t}\}$. These equilibrium objects satisfy the following conditions: households maximize their utility subject to their budget constraints, intermediate-good firms maximize the net present value of their per-period profits, final-good firms maximize profits, all of the feasibility constraints are satisfied.

¹⁵ See Matsuyama and Ushchev (2023a) and Matsuyama (2025). Note that as already shown in Eq. (5), under CES, $s(z) = \gamma_{CES} z^{1-\theta}$, where $\theta > 1$, this term is equal to $1/(\theta - 1)$.

Preference. As is customary, we impose the following functional forms to ensure the existence of the steady state:

$$u(C_t) := \frac{C_t^{1-\sigma} - 1}{1-\sigma}, v(L_t) := \frac{L_t^{1+\psi}}{1+\psi}, \quad (18)$$

where σ and ψ denote the relative risk aversion (inverse of the intertemporal elasticity of substitution) and the inverse of Frisch elasticity, respectively.

In contrast, no specific functional forms for the market share function in the HSA demand system in Eq. (1) are not necessary to ensure the steady state. Therefore, we will show analytical results without imposing any specific functional forms in Section 4. In numerical analysis provided in Sections 5 and 6, and Appendix H, parametric families of the HSA demand system are examined.

De-trending. We de-trend prices by normalizing as follows:

$$w_t := \frac{W_t}{P_t}, d_t := \frac{D_t}{P_t}, v_t := \frac{V_t}{P_t}, z_t := \frac{p_t}{A_t}, \bar{p}_t := \frac{p_t}{P_t}, \pi_t := \frac{p_t}{p_{t-1}}.$$

Notice that both P_t and A_t , i.e., the reference prices for households and firms, include entry effects. Since entry effects are not included in the official price statistics, the inflation rate is measured by PPI p_t throughout this paper. Since the equilibrium is symmetric under Rotemberg pricing, this is equal to the individual prices of intermediate inputs producers.

System of equations. Appendix A displays the system of nonlinear as well as log-linearly approximated equations in addition to the steady state conditions.

4. Competition and the Phillips curve

Now, we are ready to derive NKPC under HSA. By solving the intermediate inputs producers' problem, which is to maximize the value in Eq. (9) subject to Eqs. (11) to (10),

$$\hat{\pi}_t = \beta(1-\delta) \mathbb{E}_t \hat{\pi}_{t+1} + \frac{1}{\chi} \left[-\frac{s'(z)z}{s(z)} \right] (\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t) + \frac{1}{\chi} \left[\frac{s'(z)z}{s(z)} - \frac{\frac{s''(z)z}{s'(z)}}{1 - \frac{s(z)}{s'(z)z}} \right] \hat{z}_t, \quad (19)$$

where $\hat{x}_t := \ln(x_t/x)$ denotes the percentage deviation of x_t from the steady state value x . $\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t$ is the inverse of the markup rate, and, therefore comparable to the real marginal cost measure in the canonical New Keynesian model.¹⁶ From Eq. (16), $\hat{z}_t$ can be shown to be positively correlated with $\hat{N}_t$ as follows:

$$\hat{N}_t = - \overbrace{\frac{s'(z)z}{s(z)}}^{\ominus} \hat{z}_t \quad (20)$$

Then, using this equation, we can further simplify Eq. (19) to obtain NKPC under HSA:

$$\hat{\pi}_t = \beta(1-\delta) \mathbb{E}_t \hat{\pi}_{t+1} + \overbrace{\frac{\zeta(z)-1}{\chi}}^{\text{steady-state effect}} (\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t) - \overbrace{\frac{1}{\chi} \frac{1-\rho(z)}{\rho(z)}}^{\text{dynamic effect}} \hat{N}_t, \quad (21)$$

where

$$\zeta(z) := \frac{\partial \ln(c_t(\omega))}{\partial \ln(p_t)} = 1 - \frac{s'(z)z}{s(z)} > 1, \quad (22)$$

is the price elasticity function (hence, the markup rate under the flexible price would be $\mu^f(z) = \zeta(z)/(\zeta(z)-1)$), and

$$\rho(z) := \frac{\partial \ln(p_t)}{\partial \ln(W_t/Z_{P,t})} = \left[1 - \frac{d \ln\left(\frac{\zeta(z)}{\zeta(z)-1}\right)}{d \ln(z)} \right]^{-1}, \quad (23)$$

is the pass-through rate that prevails under the flexible price.¹⁷ We call hereafter $\rho(z)$ the pass-through rate function.

Eq. (21) contains the standard textbook NKPC under CES, such as Galí (2015), Walsh (2010) and Woodford (2003) as a special case. It is obtained by setting $s(z_t) = \gamma_{CES}(z_t)^{1-\theta}$, where θ denotes the constant price elasticity, and $\zeta(z) = \theta$ and $\rho(z) = 1$:

$$\hat{\pi}_t = \beta(1-\delta) \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\theta-1}{\chi} (\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t).$$

¹⁶ Eq. (19) expresses NKPC in terms of the real marginal cost measure. This is because, due to one-period time-to-build lag, there is an endogenous state variable in this model, the number of firms $\hat{N}_t$, hence, the real marginal cost term cannot be replaced simply by the contemporaneous output gap. In Appendix C, we consider an alternative specification with entry without one-period time-to-build lag, and derive NKPC in terms of the output gap. Even though the resulting expression is substantially more complicated than Eq. (19), the main implications do not change.

¹⁷ The pass-through rate is closely related to the super-elasticity coined by Klenow and Willis (2016). In Appendix D, NKPC is expressed in terms of the super-elasticity. Building on the insights of Baqaee et al. (2024) and Auclert et al. (2024), that highlight the usefulness of the pass-through rate in the NKPC framework, we find that expressing the NKPC in terms of the pass-through rate provides a more intuitive understanding of how competition influences the NKPC.

As is well-known, competition does not affect the Phillips curve even with endogenous entry under CES. Market concentration neither changes the slope of the Phillips curve nor exerts any direct impact on the inflation rate. The constant elasticity makes the competitive environment irrelevant to price dynamics.

By extending NKPC from CES to HSA, Eq. (21) introduces two channels through which competition affects price dynamics. First, the slope of the Phillips curve is no longer constant and determined by the steady state conditions. Thus, changes in the competitive environment will affect the slope of the Phillips curve. We call this *the steady-state effect of concentration*. Second, endogenous fluctuations in N_t and hence in z_t , affect the inflation rate directly in NKPC, which is captured by the this endogenous cost-push term. We call this effect in Eq. (21) *the dynamic effect of endogenous entry*.¹⁸

Even though HSA endogenizes the price elasticity, the reason why the slope of NKPC in Eq. (21) depends on the price elasticity is the same with the textbook NKPC under CES. Recall that under CES, the optimal price setting condition is given by

$$(1 - \theta) + \theta mc_t - \chi (\pi_t - 1)^2 \pi_t + \mathbb{E}_t \beta (1 - \delta) \frac{u'(C_{t+1})}{u'(C_t)} \chi (\pi_{t+1} - 1) \pi_{t+1} \frac{Y_{t+1}}{Y_t} = 0,$$

where mc_t denotes the real marginal cost. In response to price increase, the second term shows how much costs decline, which is proportional to the price elasticity, while the third and fourth terms represent losses and gains from price adjustments, which are expressed by inflation rates. Consequently, the higher the price elasticity, the more the demand declines and the higher the impact of the marginal cost on inflation rates becomes. In other words, with an exogenous change in the real marginal cost, the incentive to set the price closer to the target level (i.e., the price prevailing under the flexible price) becomes stronger with higher price elasticity. This motive stems from the state dependency in Rotemberg (1982) pricing and therefore is absent in Calvo (1983) pricing.

4.1. Implications of the second law

Under the Second law, $\zeta'(z) > 0$ and hence the pass-through is incomplete: $\rho(z) < 1$. Therefore, higher cost implies lower markup rates.

More concentration means higher market share $s(z)$ and lower z . The Second law, $\zeta'(z) > 0$, therefore causes a smaller slope coefficient $(\zeta(z) - 1)/\chi$ in NKPC as Eq. (21) shows.

Also, the Second law creates the endogenous cost-push term, i.e., the dynamics effect of endogenous entry. From Eq. (23), the markup rate under the flexible price is not constant:

$$\hat{\mu}_t^f = -\frac{1 - \rho(z)}{\rho(z)} \hat{z}_t = -\frac{1 - \rho(z)}{\rho(z)} (\hat{p}_t - \hat{A}_t). \quad (24)$$

The firm reduces its price and the markup rate in response to more competitive pressures, i.e., a lower A_t , when other firms reduce their prices. This is indeed strategic complementarity in price setting under HSA due to the Second law.

This endogenous cost-push term has an additional implication on aggregate fluctuations. If $\hat{\mu}_t^f = -(\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t)$ and $\hat{N}_t$ move to the opposite directions (same direction) to a structural shock, its impact on inflation rates is muted (amplified). As the analyses in Appendix H reveal, for example, to a positive technology shock, the number of firms $\hat{N}_t$ increases through the wealth effect, but the strategic complementarity in price setting reduces the markup rate $\hat{\mu}_t$ as shown in Eq. (24). As a result, the dynamic effect of endogenous entry weakens the responses of inflation rates to real economic variables.

This supply side effects of monetary policy resemble those in Baqaee et al. (2024), but our mechanism is different. In their model, monetary easing weakens the price-raising effects of monetary policy by shifting resources to more efficient firms through the misallocation channel. In our model, on the other hand, monetary policy influences the inflation rate through the number of firms under the Second law, in addition to the conventional mechanism of aggregate demand control.

Omitted variable bias. Under the Second law, competition affects price setting dynamically through strategic complementarity, as captured by the endogenous cost-push term. The imperfect pass-through leads firms to lower its price and the markup rate in response to higher competitive pressures. Thus, a naive regression of the inflation rate on the real marginal cost results in the omitted variable bias.

Appendix I shows that the endogenous cost-push term leads to the negative omitted variables bias, which is amplified with more concentration. Under the Second law, the omitted variable bias leads to the underestimation of the slope coefficient of NKPC and concentration magnifies this negative omitted variable bias. Under Rotemberg pricing, the Third law matters for the magnitude of the negative bias. Under Calvo pricing, both the Second and the Third laws matter for the magnitude of the negative bias. As a result, market concentration also weakens the observed relationship between the inflation rate and the marginal cost, thereby amplifying the supply side effects of monetary policy.

4.2. Cyclicalities of the markup rate

Table 2 summarizes the cyclicalities of the markup rate to the positive technology shock.

¹⁸ For the expositional purpose, we use the log-linearized versions of NKPC around different steady states in the main text, because log-linearization allows us to decompose the total effects of endogenous entry into the steady-state and the dynamic effects. Such a decomposition would not be possible under the nonlinear NKPC derived as equation (32) in Appendix A.

Table 2
Cyclicalities of the markup rate to the technology shock.

	Flexible price	Sticky price
CES	constant	procyclical
the Second law	countercyclical	ambiguous

In a flexible price equilibrium under CES, the markup rate is constant. In a sticky price equilibrium under CES, the markup rate becomes procyclical. This is because the marginal cost decreases but the price does not change, at least, for a short-run.¹⁹ In a flexible price equilibrium under the Second law, the markup rate becomes countercyclical because a positive technology shock increases the number of firms (varieties), which causes the markup rate to decline.

This explains why the cyclicalities of the markup rate in a sticky price equilibrium under the Second and the Third laws is generally ambiguous and depends on the tension between nominal rigidities and the pass-through rate. This can be seen by rewriting Eq. (21) in terms of the markup rate under the sticky price:

$$\hat{\mu}_t = \frac{1}{\zeta(z) - 1} \left\{ \overbrace{\chi \left[\beta (1 - \delta) \mathbb{E}_t \hat{\pi}_{t+1} - \pi_t \right]}^{\text{nominal rigidity}} \quad \overbrace{- \frac{1 - \rho(z)}{\rho(z)} \hat{N}_t}^{\text{incomplete pass-through}} \right\}. \quad (25)$$

As the pass-through rate becomes larger (smaller) and/or prices become stickier (more flexible), the markup rate becomes more procyclical (countercyclical). This could explain the disagreement about the cyclicalities of the markup in the literature as surveyed by [Nekarda and Ramey \(2020\)](#).

4.3. Why HSA?

The readers may wonder why we use HSA, in particular, in what sense HSA is critical for the way market concentration affects the shape of the Phillips curve in Eq. (21), as we discussed in this section. First of all, our results do not hold under general homothetic demand systems. As pointed out in the recent survey article on non-CES demand systems, [Matsuyama \(2025\)](#), the class of homothetic demand systems is just so broad that *almost anything goes*. For example, assuming the Second law does not guarantee procompetitive entry. In other words, even if we assume the Second law, the reduction in the number of firms caused by an increase in entry cost could lead to lower equilibrium markup rate. One advantage of HSA is that the Second law guarantees procompetitive entry, and hence restrictive entry leads to higher markup rate.

Of course, HSA is not the only sub-class of homothetic demand systems, in which the Second law guarantees procompetitive entry. This property also holds under HDIA (Homothetic with Direct Implicit Additivity), which is effectively the Kimball aggregator extended to allow for an endogenous number of firms/products, and HIIA (Homothetic with Indirect Implicit Additivity), the two alternative homothetic demand systems proposed by [Matsuyama and Ushchev \(2017, 2020b\)](#). Moreover, as shown in Appendix E, the Phillips curves under the three classes, HSA, HDIA and HIIA, are isomorphic. However, it would be challenging to characterize the general equilibrium effect of the entry cost under HDIA and HIIA, particularly the dynamic response of the number of firms, and hence those of markup and pass-through rates. It is not even straightforward to guarantee the existence and the uniqueness of equilibrium with endogenous entry. This is because, unlike HSA, it is necessary to keep track of two aggregators under HDIA and HIIA to solve for market share of each product and hence general equilibrium.²⁰ For this tractability reason, we use HSA instead of HDIA and HIIA.

5. Steady state analysis

We now simulate the New Keynesian model under HSA. In this section, we solve the non-stochastic, non-linear model presented in Appendix C.5, varying the degree of market concentration, in order to examine how changes in market concentration affect the equilibrium and, in turn, the slope of the Phillips curve.

We use three parametric families of HSA: CES, Translog and Co-PaTh. Co-PaTh is a parametric family of the HSA demand system proposed by [Matsuyama and Ushchev \(2020a\)](#). This parametric family is characterized by the property that under the flexible price equilibrium, the pass-through rate is given by a constant parameter between zero and unity. [Table 3](#) shows the market share function, the price elasticity, and the pass-through rate for each family.

¹⁹ The procyclical response of markups is quantitatively moderate. This indicates that the CES framework imposes rigidity on firms' desired markups, thereby constraining their sensitivity to cyclical fluctuations in economic activities.

²⁰ This statement does not contradict with the earlier statement – The Phillips curves under HSA, HDIA, and HIIA are isomorphic with one another –, because the latter is due to the property that, under all three classes, the price elasticity and the pass-through rate for each product are both functions of a single variable. For details, see [Matsuyama \(2025\)](#).

Table 3
Parametric families of HSA.

	Market share	Price elasticity	Pass-through
CES	$s(z) = \gamma_{CES} z^{1-\theta}$	$\zeta(z) = \theta$	$\rho(z) = 1$
Translog	$s(z) = \gamma_{TL} \ln\left(\frac{z}{z}\right)$	$\zeta(z) = 1 + \frac{1}{\ln\left(\frac{z}{z}\right)}$	$\rho(z) = \frac{1+\ln\left(\frac{z}{z}\right)}{2+\ln\left(\frac{z}{z}\right)}$
Co-PaTh	$s(z) = \gamma_{CP} \theta^{\frac{1}{1-\rho}} \left[1 - \left(\frac{z}{z}\right)^{\frac{1-\rho}{\rho}}\right]^{\frac{\rho}{1-\rho}}$	$\zeta(z) = \frac{1}{1 - \left(\frac{z}{z}\right)^{\frac{1-\rho}{\rho}}}$	$\rho(z) = \rho < 1$

Table 4
Calibrated parameters.

Parameter	Definition	Value
β	subjective discount factor	0.99
σ	relative risk aversion	1
δ	exit rate	0.025
ψ	inverse of labor supply elasticity	1
f_E, Z_E, Z_P	technologies	1
θ	price elasticity under CES	3.8
χ	Rotemberg adj. cost	77
α_i	policy inertia	0.9
α_π	policy reaction to π	1.1 or ∞
ρ	pass-through rate	1, 0.9 or 0.5

We set $\bar{z} = (\theta/(\theta - 1))^{\rho/(1-\rho)}$ so that as $\rho \rightarrow 1$, Co-PaTh converges to CES. Even though both Translog and Co-PaTh have the choke price, the inequality constraint that the price must be lower than the choke price need not be taken into account. This is because firms are symmetric; The demand for all firms, i.e., the aggregate demand, can never be zero in equilibrium.

Under CES, the price elasticity is constant and, hence, the markup rate is constant and the pass-through rate is unity. Under both Translog and Co-PaTh, the price elasticity is increasing in z . Thus, both satisfy the Second law.

Parameters are calibrated as in Table 4. Most of them are taken from Bilbiie et al. (2014). When the Taylor rule coefficient is infinity, namely, $\alpha_\pi = \infty$, the model replicates the flexible price equilibrium. We examine several pass-through rates. We set $\bar{K}$ so that $p_t/A_t = z_t = \bar{p}_t = p_t/P_t$ under CES. From Eq. (5), this can be achieved by setting $\bar{K} = 1/(\theta - 1)$.²¹ γ_{TL} and γ_{CP} in Table 3 are calibrated so that $\zeta(z) = \theta$ when $f_E = 1$.²² γ_{TL} and γ_{CP} affect z in the steady state.

In Fig. 1, we replicate higher market concentration, and the rising profit margins and markups reported by Covarrubias et al. (2019), De Loecker et al. (2020), and Autor et al. (2020). The horizontal axis displays the entry cost; the further to the right, the greater the barriers to entry. The green line corresponds to CES, the red line to Translog, and the blue lines to Co-PaTh, when $\rho = 0.9$ (dotted) and $\rho = 0.5$ (dashed), respectively. In all cases, an increase in the entry cost leads to market concentration (decline in the number of firms). Under Translog and Co-PaTh, this also leads to higher markup rates and profits.²³ The Second law embedded in Translog and Co-PaTh enables the model to replicate the stylized facts as observed in the data. Notice also that Translog is similar to Co-PaTh with $\rho = 0.5$.

Fig. 2 illustrates how market concentration affects the slope of the Phillips curve. The horizontal axis shows the entry cost in the upper panel. There, the greater the barriers to entry, the flatter the Phillips curve. The lower panel displays how the slope of the Phillips curve changes with the number of firms. Since the greater the cost of entry, the fewer the number of firms, the lower panel is just like a mirror image of the upper panel. Fewer firms in the market result in the flattening of the Phillips curve.

Fewer firms (higher market share) corresponds to lower z , as shown in Eq. (20). Under the Second law, $\zeta'(z) > 0$, this leads to lower price elasticity and consequently, the slope of the Phillips curve $(\zeta(z) - 1)/\chi$ declines with higher market concentration.

The slope of the Phillips declines more to market concentration as the pass-through rate becomes smaller. This is because, as implied in Eq. (24), the smaller pass-through rate causes the markup rate and therefore the price elasticity to react more strongly to competitive pressures.

De Loecker et al. (2020) argue that “aggregate markups start to rise from 21% above marginal cost to 61% now.” This increase in the markup rate of about 40 percentage points suggests that the entry cost would have increased by about 3.5 times in Translog, and by about 2.5 times in Co-PaTh with $\rho = 0.5$ in Fig. 1. According to Fig. 2, the accompanying market concentration can halve the slope of the Phillips curve. The Second law is a well-established fact provided by many empirical studies. Thus, it could be said that changes in the competitive environment in recent years have exerted a significant impact on the slope of the Phillips curve.

Appendix H describes how the economy under the HSA demand system reacts to technology and monetary policy shocks. Since market concentration generates a sizable reduction in the slope of NKPC, the responses of the inflation rate to both technology and monetary policy shocks become smaller as market concentration deepens in impulse responses.

²¹ This is equivalent to setting $Z_C = \gamma_{CES}^{1/(\theta-1)}$.

²² Price elasticity is given by θ under CES and thus determined independently from γ_{CES} . We set γ_{CES} at the same value of γ_{CP} when $\rho = 0.5$.

²³ The bottom right panel in Fig. 1 shows a nonlinear relationship between the entry cost and the profit in the steady state. This is because the entry cost entails wage payments, as shown in Eq. (13).

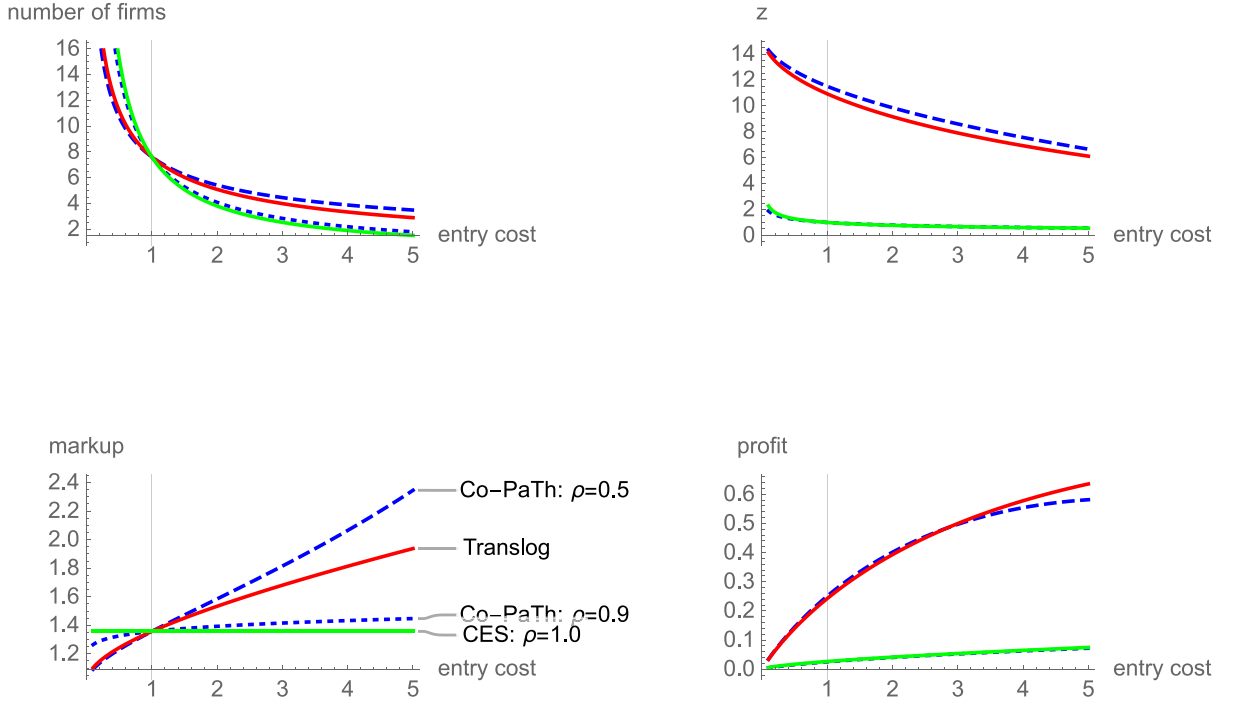


Fig. 1. Concentration and macroeconomy.

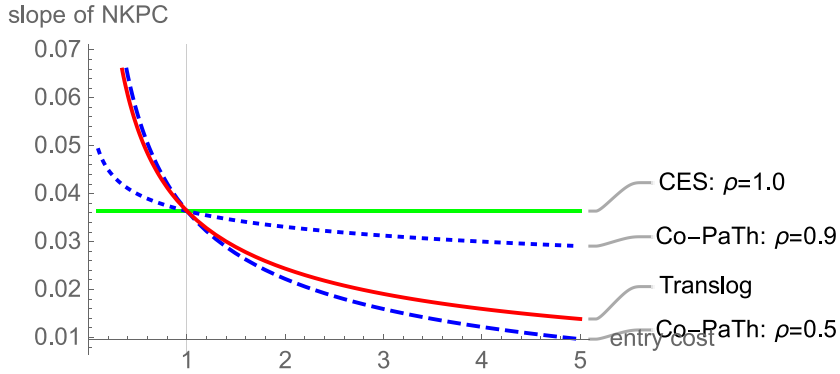


Fig. 2. Entry cost, concentration and the slope of the Phillips curve.

6. Calvo pricing

So far, we have used the Rotemberg (1982) pricing as the source of sticky prices. This helps to highlight the role of the Second law due to its state dependency and symmetric pricing, by shutting down the role of the Third law. We now consider Calvo (1983) pricing for its heterogeneous pricing and time dependency, which shut down the role of the Second law. This helps to highlight the role of the Third law on the flattening of the Phillips curve. Details of the derivation are given in Appendix F.

NKPC under HSA and Calvo pricing is given by

$$\hat{\pi}_t = \beta(1-\delta)\mathbb{E}_t\hat{\pi}_{t+1} + \frac{(1-\phi)[1-\phi\beta(1-\delta)]}{\phi} [\rho(\bar{z})(\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t) - (1-\rho(\bar{z}))\hat{N}_t], \quad (26)$$

where $1-\phi$ is the probability that a firm can adjust its price. Under Calvo pricing, unlike Rotemberg pricing, not all firms set the same prices. Therefore, we define the average price under Calvo friction denoted by $\bar{p}_t$, implicitly given by

$$\int_{\Omega_t} s\left(\frac{p_t(\omega)}{A_t}\right) d\omega = 1 = N_t s\left(\frac{\bar{p}_t}{A_t}\right).$$

Accordingly, $\bar{z}_t := \bar{p}_t/A_t$ and $\hat{\pi}_t := \bar{p}_t/\bar{p}_{t-1}$.

Now compare the slope of the two Phillips curves under Rotemberg and under Calvo. Notice that the causal impact from market concentration to the flattening of the Phillips curve is summarized by the price elasticity $\zeta(z)$ under Rotemberg in Eq. (21), and by the pass-through rate $\rho(z)$ under Calvo in Eq. (26). Recall that the Second law implies $\zeta'(z) > 0$ and the Third law implies $\rho'(z) > 0$.²⁴ Thus, what creates the causal link from market concentration to the flattening of the Phillips curve is the Second law, not the Third law, under Rotemberg. In contrast, it is the Third law, not the Second law, under Calvo.

Why the Second law, not the Third law, matters for Rotemberg, while the Third law, not the Second law, matters for Calvo. Under Rotemberg, all firms remain symmetric and adjust their prices simultaneously, so a common cost shock leaves relative prices unchanged, implying that the pass-through rate, hence the Third law, plays no role on pricing decisions. Moreover, the common price can move only gradually, due to the price adjustment cost, which is increasing in the adjustment speed. This makes the firms choose their adjustment speed based on their desire to minimize the profit loss caused by the discrepancies between the current markup rate and the target markup rate. This is why demand elasticity, and hence the Second law, matters. In contrast, under Calvo, the firms that can adjust their prices are not subject to the price adjustment cost, so that they can immediately reset their prices to achieve their target markup rate. This is why the demand elasticity, and hence the Second law, does not matter. However, the firms that can adjust prices need to take into account the fact that not all firms adjust their prices when calculating their target markup rate. This makes the pass-through rate, and hence the Third law, matter for the size of individual price adjustments, which in turn determines the speed of the aggregate price movement.

We would like to stress that by showing the irrelevance of the Third law under Rotemberg, we do not mean to suggest that the Third law is unimportant. Likewise, by showing the irrelevance of the Second law under Calvo, we do not mean to suggest that the Second law is unimportant. On the contrary, we believe both are important. But Rotemberg helps to isolate the role of the Second law, because its symmetric pricing shuts down the effect of the Third law, and Calvo helps to isolate the role of the Third law, because its lack of the state dependency shuts down the effect of the Second law.²⁵

6.1. Power elasticity of markup rate (PEM)

Co-PaTh satisfies the Third law only in the weak form: the pass-through rate is constant. PEM, another parametric family of HSA, satisfies the Third law in the strong form: a higher price leads to a smaller rate of change in the price elasticity.²⁶

PEM is defined by the following market share function:

$$s(z) = \exp \left[\int_{z_0}^z \frac{c}{c - \exp \left(-\frac{\kappa \bar{z}^{-\lambda}}{\lambda} \right) \exp \left(\frac{\kappa \xi^{-\lambda}}{\lambda} \right)} \frac{d\xi}{\xi} \right], \quad (27)$$

with either $\bar{z} = \infty$ and $c \leq 1$ or $\bar{z} < \infty$ and $c = 1$. By using Eq. (22), the price elasticity under PEM is given by

$$\zeta(z) = \frac{1}{1 - c \exp \left(\frac{\kappa \bar{z}^{-\lambda}}{\lambda} \right) \exp \left(-\frac{\kappa z^{-\lambda}}{\lambda} \right)}, \quad (28)$$

and using Eq. (23), the pass-through rate under PEM is given by

$$\rho(z) = \frac{1}{1 + \kappa z^{-\lambda}}. \quad (29)$$

Clearly, $\kappa > 0$ and $\lambda > 0$ ensure both the Second law and the Third law in the strong form.

Notice that by comparing Eqs. (23) and (29), one can immediately see

$$-\frac{d \ln \left(\frac{\zeta(z)}{\zeta(z)-1} \right)}{d \ln(z)} = \kappa z^{-\lambda},$$

which means that the elasticity of the markup rate under the flexible price is a power function of z . This is why Matsuyama and Ushchev (2023b) call this family Power Elasticity of Markup Rate.²⁷ PEM contains Co-PaTh and CES as limit cases; It collapses to Co-PaTh with $\lambda = 0$, $\kappa = (1 - \rho)/\rho$, $c = 1$ and $\bar{z} < \infty$, and to CES with $\kappa = 0$, $c = 1 - 1/\theta$ and $\bar{z} = \infty$. When PEM collapses to CES, the market share function is given by $s(z) = (z/z_0)^{1-\theta}$. To be comparable between CES and PEM, we set $z_0 = \gamma^{\frac{1}{\theta-1}}$.²⁸

²⁴ Notice that as evident in Table 3, Translog violates the Third law.

²⁵ To confirm this intuition, we also rederive NKPC under Taylor (1980) contract, which features heterogenous pricing and its lack of state dependency, to demonstrate that it is the Third law, not the Second law, that matters: equation (54) in Appendix G.

²⁶ For more details, see Appendix D.3. in Matsuyama and Ushchev (2023b).

²⁷ Matsuyama and Ushchev (2023b) also call this family Fréchet Inverse Markup Rate, because its inverse markup rate function is proportional to the familiar Fréchet distribution function:

$$1 - \frac{1}{\zeta(z)} = c \exp \left(\frac{\kappa \bar{z}^{-\lambda}}{\lambda} \right) \exp \left(-\frac{\kappa z^{-\lambda}}{\lambda} \right).$$

²⁸ γ is set at a value so that $\zeta(z) = \theta$ when $f_E = 1$ and $\rho = 0.5$. Notice that γ does not affect the price elasticity and the pass-through rate as evident in Eqs. (28) and (29).

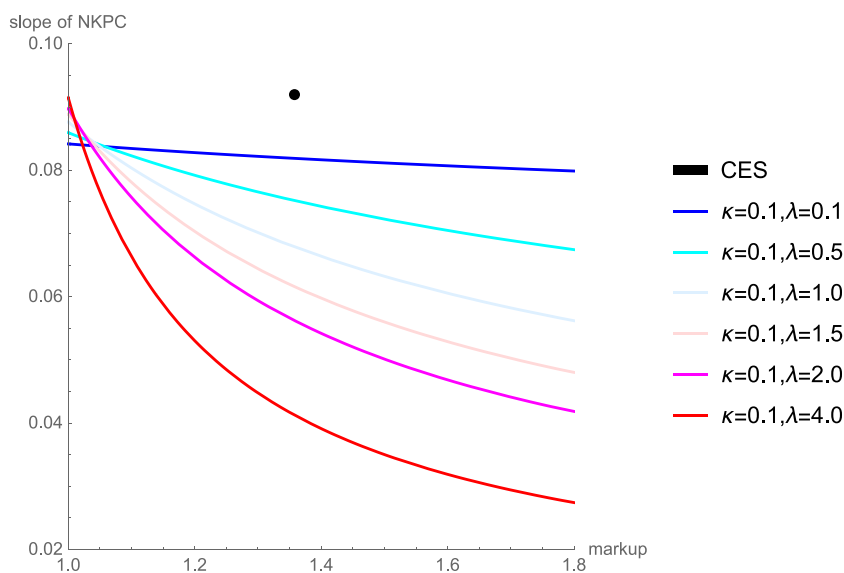


Fig. 3. Markup and the slope of the Phillips curve under Calvo pricing.

In Fig. 3, each locus traces the markup rate $\zeta(\bar{z}) / [\zeta(\bar{z}) - 1]$ and the slope of NKPC $(1 - \phi)[1 - \phi\beta(1 - \delta)]\rho(\bar{z})/\phi$ in Eq. (26) by changing $\bar{z}$ under several λ with the reset probability $1 - \phi = 0.25$.²⁹ The black dot indicates the slope of the Phillips curve and the steady state markup under CES. There, the price elasticity, and therefore the markup rate, is not affected by competitive pressures, so the markup rate is $\theta/(\theta - 1)$ and the slope of the Phillips curve is $(1 - \phi)[1 - \phi\beta(1 - \delta)]/\phi$. Eq. (29) shows that as λ approaches to zero, the pass-through rate hardly moves, *i.e.*, PEM collapses to Co-PaTh. Therefore, variations in the pass-through rate become smaller. λ controls how variable the pass-through rate is to changes in z as implied in Eq. (29), while κ determines the level of the pass-through rate.

According to De Loecker et al. (2020), the markup rate increased to 21% to 61% in recent years. Although it is difficult to draw quantitative conclusions since there are no estimated values for κ and λ , Fig. 3 shows that even under Calvo pricing, changes in the competitive environment among firms can result in a much smaller slope of the Phillips curve through the Third law.

7. Conclusion

To understand the causal relationship from market concentration to the flattening of the Phillips curve, this paper extended the canonical New Keynesian model by replacing CES with HSA to allow endogenous entry to cause markup rate and pass-through rate changes. HSA demand system is well-suited for this purpose because the market share function and its single aggregator fully summarize all the information required to understand how market concentration affects the markup rate and the pass-through rate.

It has been shown that a higher entry cost, which leads to market concentration, causes the flattening of the Phillips curve for two reasons. First, market concentration *structurally* reduces the slope of the Phillips curve, by causing a rise of the markup rate (under the Second law) and a decline of the pass-through rate (under the Third law). We demonstrated the former under Rotemberg pricing and the latter under Calvo pricing. Second, due to the endogenous cost-push shock, *i.e.*, the supply side effects of monetary policy, emerging under the Second law, a *naïve* regression of the real marginal cost on the inflation rate leads to the negative omitted variables bias, which is amplified by market concentration under the Second and the Third laws. This weakens the *estimated* relationship between the inflation rate and the marginal cost. These two reasons, one structural and one observational, together can go a long way toward understanding the flattening of the Phillips curve.

In this paper, we utilized two canonical pricing mechanisms: Rotemberg and Calvo. Rotemberg pricing highlights the effect of the Second law on the slope of the Phillips curve due to its state dependency and symmetric pricing, which shuts down the role of the Third law. In contrast, Calvo pricing highlights the effect of the Third law on the slope of the Phillips curve due to its heterogeneous pricing and time dependency, which shuts down the role of the Second law. In a more general pricing mechanism, such as the menu cost, which features both state and time dependencies as well as heterogeneous pricing, we conjecture that the endogeneity of both the markup rate and the pass-through rate should be important and that the results would be a hybrid of Rotemberg and Calvo. Of course, a menu cost model would be necessary for assessing numerically the relative importance of the two, which is left for future research. Other research topics that could be addressed by applying HSA to the New Keynesian framework may include optimal monetary policy, the wage Phillips curve, and heterogeneous productivity across firms.

²⁹ As in Fig. 2 in Section 4, it is also possible to consider this as a case in which the entry cost changes.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmoneco.2026.103920>.

Data availability

No data was used for the research described in the article.

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